

MIST: Integrating Local and Tissue-Wide Context for Gene Expression Prediction from Histology

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Abstract

Spatial transcriptomics reveals how gene expression is organized across a tissue, but it is costly and slow, motivating methods that predict it directly from routine H&E histology. Existing predictors, however, are typically trained per dataset, are sensitive to the arbitrary coordinate frame of each slide, or rely on slow iterative generative sampling. We introduce MIST, a compact spatial transformer that predicts a slide’s spatial gene expression in a single forward pass by fusing three complementary views of context in every layer: local attention over spatial nearest neighbors, global self-attention across all spots, and an attention-pooled slide representation. Because spot coordinates enter only through pairwise distances, its predictions are provably invariant to translations, rotations, and reflections of the coordinate frame when image features are fixed. On six human organ cohorts from HEST-1k, evaluated under intra-cohort, pooled multi-organ, and leave-one-organ-out settings, MIST attains the highest mean Pearson correlation (PCC) on every cohort, improving over the strongest baseline by 11.6% (50 genes) and 11.4% (100 genes), and the lowest MSE and MAE; the gains are statistically significant and persist across UNI, CONCH, and combined features. MIST also preserves spatial tissue structure, recovering the expression domains of the measured data more faithfully than the strongest baseline on 78–90% of held-out slides. Beyond predictive performance,

MIST recovers canonical cancer-marker genes, reproducing the spatial expression of melanoma (mean PCC 0.90) and breast HER2/ER/PR markers, so it reads out clinically actionable molecular signals from H&E alone. It generalizes to held-out organs (+6.0% over the closest baseline BLEEP), while transfer to entirely unseen anatomy remains the main open challenge.

1 Introduction

Spatial transcriptomics (ST) measures gene expression together with tissue location, enabling the study of spatial heterogeneity, tissue organization, and cellular interactions [Ståhl et al. \(2016\)](#). Its cost and experimental burden, however, constrain the number and diversity of specimens that can be profiled. In contrast, H&E slides are routinely collected and contain morphological signals associated with tissue composition and molecular state [Chen et al. \(2024a\)](#); [Jaume et al. \(2024\)](#). Predicting spatial gene expression from these images could therefore extend molecular analysis to specimens without matched ST measurements [He et al. \(2020\)](#); [Zeng et al. \(2022\)](#); [Xie et al. \(2023\)](#).

Morphology and expression interact locally, across the slide, and at the level of slide composition. Existing predictors capture parts of this structure with patch regression, spatial graphs, retrieval, or conditional generation [He et al. \(2020\)](#); [Zeng et al. \(2022\)](#); [Xie et al. \(2023\)](#); [Zhu et al. \(2025\)](#), but absolute coordinates can make predictions depend on an arbitrary scanner frame. Geometric representations based on relative quantities provide a principled way to reduce this dependence [Satorras et al. \(2021\)](#). We therefore seek a compact predictor that combines multi-scale context while using coordinates only through relative distances.

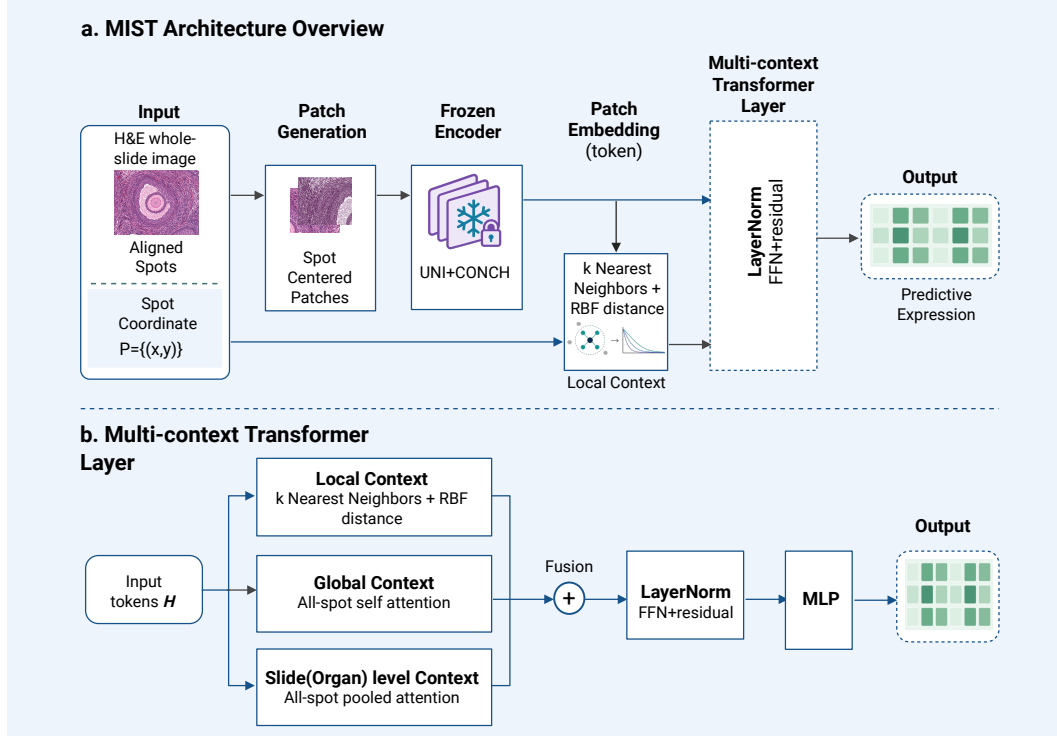


Figure 1: MIST overview. (a) Frozen UNI, CONCH, or combined features turn H&E spot patches into tokens, while coordinate-based k NN edges provide local context. (b) Each layer combines distance-biased local attention with global self-attention before the expression head. Pairwise-distance coordinates ensure invariance to translations, rotations, and reflections.

Table 1: Architectural components across representative histology to spatial transcriptomic (H&E to ST) predictors. • present, ○ absent.

Method	Local context	Global context	Slide token	Coord. invariant	Cont. distance
ST-Net He et al. (2020)	○	○	○	○	○
HisToGene Pang et al. (2021)	○	•	○	○	○
Hist2ST Zeng et al. (2022)	•	•	○	○	○
BLEEP Xie et al. (2023)	○	○	○	○	○
TRIPLEX Chung et al. (2024)	•	•	○	○	○
M2OST Wang et al. (2025)	•	○	○	○	○
STEM Zhu et al. (2025)	○	•	○	○	○
MIST (ours)	•	•	•	•	•

We introduce MIST, a direct spatial transformer that represents each spot as a token and combines spatial k NN attention for local context, full self-attention for slide-wide context, and attention pooling for a slide summary [Vaswani et al. \(2017\)](#). Because spot coordinates enter the model only through pairwise distances, its predictions are invariant to translations, rotations, and reflections when image features are fixed (Fig. 1).

We evaluate MIST across six human cohorts under intra-cohort prediction, pooled multi-organ prediction, and leave-one-organ-out transfer. In intra-cohort evaluation, MIST achieves the highest mean Pearson correlation and the lowest mean squared error (MSE) and mean absolute error (MAE), averaged across cohorts, for all evaluated gene-selection strategies and panel sizes. Pooled and leave-one-organ-out evaluations further assess generalization across organs, highlighting the challenges of transfer to unseen anatomy.

Our experiments address four research questions: RQ1 (Performance): How accurately does MIST predict gene expression across organs when trained separately per cohort or jointly across cohorts? RQ2 (Transfer): How well does it generalize to an organ excluded from training? RQ3 (Design): How do the image encoder, context pathways, and neighborhood construction affect predictive accuracy? RQ4 (Structure): How well do the predicted expression profiles preserve spatial domains identified from measured expression?

Contributions.

1. We introduce MIST, a multi-scale transformer with local, slide-wide, and slide-summary pathways driven by continuous pairwise-distance features and coordinate-frame invariance.
2. We quantify the effects of context, distance encoding, and feature choice through matched ablations and transformation tests.
3. We provide a controlled pooled cross-organ benchmark alongside intra-cohort and held-out-organ evaluation with matched panels, splits, baselines, and model selection.

2 Related Work

H&E to ST prediction. Prior models span patch-level regression (ST-Net [He et al. \(2020\)](#)), graph spatial context (Hist2ST [Zeng et al. \(2022\)](#)), cross-modal retrieval (BLEEP [Xie et al. \(2023\)](#)), and conditional generation (STEM [Zhu et al. \(2025\)](#)). Because they differ in features, gene panels, splits, and selection, published scores conflate data choices with architecture; we therefore compare under a matched protocol. MIST is a deterministic point predictor and does not model the full conditional expression distribution.

Geometry and generalization. Absolute position encodings can change when an equivalent tissue layout is translated, rotated, or reflected; geometric networks instead build interactions from relative quantities [Satorras et al. \(2021\)](#). MIST conditions its spatial pathway only on pairwise Euclidean distances, invariant to such frame changes for fixed histology features. For generalization we separate pooled cross-validation (organs seen, samples held out) from LOOO (an entire organ withheld) to isolate anatomical transfer on HEST-1k [Jaume et al. \(2024\)](#). An extended discussion and positioning of MIST are in Appendix B.

3 Method

MIST maps frozen spot-level histology features and measured spot coordinates to a spatial gene expression map. Its method has three stages: coordinate-invariant distance encoding, a shared multi-context layer, and supervised prediction with a correlation-aware objective.

3.1 Problem Setup

For a slide with N spots, let $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times d_{\text{in}}}$ denote frozen pathology features and $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$ the corresponding spot coordinates. The target $\mathbf{Y} \in \mathbb{R}^{N \times G}$ contains log-normalized expression for G genes. We learn $f_\theta : (\mathbf{F}, \mathbf{P}) \mapsto \hat{\mathbf{Y}}$ and process each slide independently; the model therefore supports a variable number of spots and does not require expression measurements at test time. In the feature-matched main comparisons, $\mathbf{f}_i$ concatenates frozen UNI [Chen et al. \(2024b\)](#) and CONCH [Lu et al. \(2024\)](#) embeddings ($d_{\text{in}} = 1536$).

3.2 Coordinate-Invariant Distance Encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i , we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbors, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\phi(D_{ij}) = [\exp(-\gamma_m D_{ij}^2)]_{m=1}^M, \quad (1)$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

Proposition 1 (Coordinate frame invariance):

Let $R \in \mathbb{R}^{2 \times 2}$ be orthogonal, let $\mathbf{t} \in \mathbb{R}^2$, and define $\mathbf{P}' = \mathbf{P}R^\top + \mathbf{1t}^\top$. For fixed histology features $\mathbf{F}$, the prediction is unchanged:

$$f_\theta(\mathbf{F}, \mathbf{P}') = f_\theta(\mathbf{F}, \mathbf{P}). \quad (2)$$

Thus, the spatial pathway is invariant to translations, rotations, and reflections of the coordinate frame in exact arithmetic. The guarantee concerns a change of coordinate frame with fixed histology features; it does not claim that the frozen image encoder is invariant to transformations of the histology image. The proof is in Appendix ???. The RBF bias is available to weight neighbors by distance, but at native HEST pixel scale the neighbor distances ($\sim 10^3$) saturate $\exp(-\gamma D^2)$ toward zero for all bandwidths, so in the reported configuration the bias contributes negligibly and the local pathway reduces to attention over the spatial k NN neighborhood. The distinguishing property of MIST is therefore this spatial locality (each spot aggregates from its geometric neighbors rather than from arbitrary or random spots), which our ablation isolates directly (spatial k NN vs. random neighbors, §5.3). The invariance guarantee above holds regardless, since it depends only on distances being rotation-invariant.

3.3 Multi-Context MIST Layer

Core Idea. At every layer, each spot receives three messages in parallel: one from its nearby spots, one from all spots, and one from a pooled summary of the slide. The model adds these messages to the current spot representation and refines the result with a feed-forward network. Only the nearby-spot message uses coordinates, and it uses pairwise distances rather than absolute positions.

The input features are projected to hidden width d , $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbor graph and radial-basis features are reused in all L transformer layers. At layer ℓ , a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

3.3.1 Local distance-biased attention

Each spot attends to its k neighbors with a learned, distance-dependent bias for each attention head. For head h , the attention weights are

$$\alpha_{ij}^h = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij})) \right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query-key term, while spatial distance contributes through the learned radial-basis bias.

3.3.2 Global self-attention

In parallel, standard multi-head self-attention Vaswani et al. (2017) allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

3.3.3 Attention-pooled slide context

A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^N a_i \mathbf{h}_i \right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide’s spot features and summarizes its morphological composition.

For the complete model, the local ($\mathbf{L}$), global ($\mathbf{G}$), and broadcast slide ($\mathbf{S}$) representations are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\tilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{G}^{(\ell)} + \mathbf{S}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \tilde{\mathbf{X}}^{(\ell)} + \text{FFN}\left(\text{LN}(\tilde{\mathbf{X}}^{(\ell)})\right). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

Proposition 2 (Permutation equivariance):

Let $Q \in \{0, 1\}^{N \times N}$ be a permutation matrix. If the k NN construction handles distance ties equivariantly, then

$$f_{\theta}(Q\mathbf{F}, Q\mathbf{P}) = Qf_{\theta}(\mathbf{F}, \mathbf{P}). \quad (7)$$

Therefore, reordering the input spots only reorders their predicted expression profiles. Appendix ?? provides the proof. Our numerical checks evaluate both Propositions 1 and 2 under finite precision.

3.4 Training Objective and Ablations

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \quad (8)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

The main model enables all three context pathways. Our factorial ablation independently toggles the local (L), global (G), and slide-level (S) pathways, avoiding conclusions tied to a particular order of architectural additions.

4 Experimental Setup

We evaluate on HEST-1k [Jaume et al. \(2024\)](#) with frozen UNI+CONCH patch features (Appendix D ablates each encoder in isolation) and $\log(1+x)$ targets; unless stated otherwise the target is the 50-gene HEST HVG panel. To test robustness to how the target genes are chosen, we use three 50-gene selection strategies: HVG, the highest-variance genes; DEG, differentially expressed genes selected by a Wilcoxon test across Leiden clusters; and HMHVG, a mean-and-variability criterion applied after variance filtering (Table 2). Cross-organ experiments use the six retained cohorts of Appendix A (HCC, LYMPH_IDC, and READ excluded). We study three regimes: intra-cohort (a model per organ), pooled (train across organs, hold out slide-level samples; organ identity is not a model input), and LOOO (train on all-but-one organ, test the held-out organ). Evaluation metrics. Our primary metric is the Pearson correlation coefficient (PCC) computed per gene between predicted and measured expression and averaged over genes (genes with zero target variance are excluded). We also report the mean squared error (MSE) and mean absolute error (MAE) to capture prediction scale, and, for the spatial-domain-recovery analysis, the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) between domains clustered from predicted versus measured expression; efficiency is measured by inference latency and peak GPU memory. All metrics are averaged over outer folds and three seeds, with checkpoints

chosen on inner validation and evaluated once on the outer fold. Baselines ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) use matched features, panels, splits, budgets, and selection. MIST has 4.88M parameters (four layers, width 256, eight neighbors). Full dataset, gene-panel, protocol, and hyperparameter details are in Appendix C.

5 Results

5.1 RQ1: Predictive performance across diverse organs

We begin by asking whether MIST predicts spatial expression accurately on organs it has seen during training. To answer this, we measure accuracy in two complementary ways under matched features, panels, and splits: a separate model trained per cohort (intra-cohort), and a single model trained jointly across all organs (pooled).

Starting with the intra-cohort setting, we stress-test accuracy along two axes of the target panel and report both. First, across the three 50-gene selection strategies (Table 2), MIST attains the best mean PCC under every strategy: 0.496 (HVG), 0.615 (DEG), and 0.650 (HMHVG), each ahead of the strongest baseline (BLEEP at 0.444, 0.559, and 0.587, respectively). Second, across panel sizes (Table 3), this lead persists at every size, as MIST tops the average at PCC@10 (0.600), PCC@50 (0.496), and PCC@100 (0.559). Focusing on the default PCC@50 panel, the gain over BLEEP is 11.6%, and MIST wins all six cohorts, with the margin widest on kidney and prostate (+0.072, +0.064) and narrowest on lung (+0.024); at the smallest 10-gene panel, correlations rise across all methods because the retained genes are the most variable, and hence the most predictable. Crucially, these gains are not an artifact of the correlation metric: as Table 4 reports, MIST also achieves the lowest error under every strategy and size (for example, at PCC@50 its MSE is 1.233 and MAE 0.794, versus 1.318 and 0.819 for BLEEP). These intra-cohort gains are statistically significant, not noise: across the six cohorts and three seeds ($n=18$ paired runs), MIST’s mean PCC@50 exceeds BLEEP’s by 0.051 (95% bootstrap CI [0.041, 0.062]; Wilcoxon signed-rank $p=7.6 \times 10^{-6}$, paired t -test $p=4.7 \times 10^{-8}$), and at the finer per-gene level MIST outperforms BLEEP on 87% of genes ($n=300$; Wilcoxon $p < 10^{-30}$). Appendix ?? reports the full paired tests for all regimes, including the significant LOOO gain and the marginal pooled one.

Table 4: Intra-cohort error metrics (MSE, MAE) on HEST-1k. Mean over three seeds and six cohorts, matched protocol; best per row in bold.

Section	Panel	MSE ↓					MAE ↓				
		ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
Gene-selection strategy	HVG	1.995	1.561	1.318	1.747	1.233	0.969	0.890	0.819	0.920	0.794
	DEG	3.336	1.757	1.557	1.958	1.477	1.457	1.021	0.945	1.054	0.928
	HMHVG	3.589	1.832	1.566	1.976	1.393	1.470	1.042	0.936	1.049	0.901
Gene-panel size	10	6.087	3.297	2.859	3.181	2.620	1.908	1.374	1.243	1.310	1.211
	50	1.995	1.561	1.318	1.747	1.233	0.969	0.890	0.819	0.920	0.794
	100	3.320	1.843	1.558	1.899	1.458	1.354	1.011	0.915	1.001	0.894

Turning next to the pooled setting, we test whether a single shared model can replace the per-organ ones. Trained jointly on all organs (leave-slide-out cross-validation; no per-organ specialisation and no organ labels), one MIST predicts across the diverse organs at once, adapting to each slide through its shared local, global, and slide-level context rather than a dedicated per-tissue model. Even so, it remains the best method: on the six-organ subset it attains the top mean PCC (0.599; Table 5), ahead of BLEEP (0.587) and Hist2ST (0.556), and reaches 0.599 on the six-organ HEST panel (BLEEP 0.587). Together, these two settings answer RQ1: MIST is the most accurate predictor on seen organs, whether specialised per organ or shared across all of them, so a separate model per tissue is not required.

Table 2: Intra-cohort PCC across 50-gene selection strategies. Mean_{std} over three seeds.

Panel	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
HVG	IDC	0.551±0.01	0.305±0.11	0.519±0.00	0.434±0.00	0.581±0.01
	CCRCC	0.163±0.02	0.180±0.03	0.175±0.02	0.118±0.01	0.252±0.01
	LUNG	0.521±0.01	0.430±0.05	0.565±0.01	0.407±0.00	0.589±0.01
	PAAD	0.366±0.03	0.248±0.04	0.412±0.03	0.233±0.02	0.437±0.03
	PRAD	0.287±0.02	0.323±0.01	0.346±0.01	0.207±0.02	0.410±0.01
	SKCM	0.661±0.01	0.523±0.00	0.650±0.00	0.461±0.02	0.707±0.01
	Average	0.425	0.335	0.444	0.310	0.496
DEG	IDC	0.697±0.01	0.549±0.07	0.731±0.01	0.684±0.02	0.774±0.01
	CCRCC	0.312±0.01	0.335±0.01	0.304±0.00	0.253±0.01	0.402±0.01
	LUNG	0.566±0.01	0.518±0.01	0.621±0.01	0.440±0.01	0.662±0.00
	PAAD	0.425±0.03	0.352±0.01	0.487±0.03	0.306±0.04	0.518±0.03
	PRAD	0.413±0.00	0.437±0.01	0.466±0.00	0.355±0.03	0.539±0.01
	SKCM	0.752±0.01	0.656±0.00	0.746±0.03	0.617±0.02	0.794±0.00
	Average	0.528	0.475	0.559	0.443	0.615
HMHVG	IDC	0.724±0.01	0.633±0.08	0.739±0.02	0.687±0.01	0.774±0.01
	CCRCC	0.339±0.01	0.387±0.01	0.349±0.01	0.290±0.01	0.454±0.02
	LUNG	0.568±0.00	0.598±0.00	0.661±0.00	0.517±0.01	0.693±0.00
	PAAD	0.441±0.03	0.341±0.04	0.493±0.02	0.305±0.03	0.524±0.02
	PRAD	0.452±0.00	0.477±0.01	0.432±0.04	0.337±0.02	0.555±0.02
	SKCM	0.824±0.03	0.710±0.14	0.849±0.02	0.731±0.06	0.901±0.00
	Average	0.558	0.524	0.587	0.478	0.650

Table 3: Intra-cohort PCC across gene-panel sizes. Mean_{std} over three seeds; each panel reports its average.

Metric	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
PCC@10	IDC	0.666±0.01	0.636±0.01	0.635±0.02	0.591±0.02	0.708±0.00
	CCRCC	0.231±0.01	0.273±0.02	0.231±0.02	0.170±0.00	0.322±0.01
	LUNG	0.702±0.01	0.519±0.02	0.735±0.00	0.616±0.01	0.759±0.00
	PAAD	0.511±0.04	0.377±0.07	0.536±0.03	0.393±0.04	0.519±0.05
	PRAD	0.442±0.03	0.459±0.02	0.434±0.03	0.325±0.03	0.524±0.02
	SKCM	0.728±0.00	0.471±0.04	0.697±0.01	0.598±0.01	0.769±0.01
	Average	0.547	0.456	0.545	0.449	0.600
PCC@50	IDC	0.551±0.01	0.309±0.07	0.519±0.00	0.439±0.01	0.581±0.01
	CCRCC	0.163±0.02	0.184±0.02	0.175±0.02	0.123±0.00	0.252±0.01
	LUNG	0.518±0.01	0.470±0.01	0.562±0.01	0.423±0.03	0.589±0.01
	PAAD	0.366±0.03	0.248±0.04	0.412±0.03	0.233±0.02	0.437±0.03
	PRAD	0.294±0.02	0.333±0.02	0.348±0.01	0.207±0.02	0.410±0.01
	SKCM	0.665±0.01	0.522±0.03	0.645±0.01	0.491±0.04	0.707±0.01
	Average	0.425	0.335	0.444	0.310	0.496
PCC@100	IDC	0.611±0.01	0.473±0.07	0.632±0.01	0.565±0.01	0.682±0.01
	CCRCC	0.221±0.01	0.274±0.03	0.247±0.02	0.192±0.00 [†]	0.328±0.02
	LUNG	0.595±0.00	0.434±0.03	0.659±0.01	0.498±0.03	0.688±0.00
	PAAD	0.355±0.04	0.222±0.04	0.399±0.04	0.236±0.02	0.435±0.03
	PRAD	0.327±0.01	0.341±0.03	0.377±0.02	0.240±0.03	0.450±0.00
	SKCM	0.696±0.01	0.446±0.12	0.696±0.01	0.602±0.02	0.771±0.01
	Average	0.468	0.365	0.502	0.389	0.559

Table 5: Cross-organ POOLED (six-organ subset). 5-fold slide-level holdout; mean_{std} over three seeds; best in bold.

Method	PCC ↑
ST-Net He et al. (2020) (NBE’20)	0.351 _{0.02}
Hist2ST Zeng et al. (2022) (BIB’22)	0.556 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS’23)	0.587 _{0.00}
STEM Zhu et al. (2025) (ICLR’25)	0.273 _{0.00}
MIST	0.599_{0.01}

Key Finding (RQ1): MIST gives the best accuracy across diverse organs both per organ (intra-cohort PCC@50 0.496 vs. 0.444 for BLEEP, winning all six cohorts) and as a single pooled model (0.599 six-organ HEST), so a separate model per tissue is unnecessary when the organ is represented in training.

5.2 RQ2: Cross-organ transfer to an unseen organ

Having established strong performance on seen organs, we now ask the harder question of whether MIST transfers to an organ it has never seen. To test this, we use leave-one-organ-out (LOOO) evaluation: we train on all-but-one organ and test on the held-out one, so the target anatomy is entirely absent from training. As expected, accuracy falls relative to the pooled setting of RQ1, yet MIST still transfers best: on the six-organ subset it wins every held-out organ and attains the highest mean PCC (0.422 vs. 0.398 for BLEEP; a significant gain, Wilcoxon $p=0.018$, Appendix ??). Figure 2 illustrates this for four representative held-out organs (kidney and prostate are omitted for readability), where MIST (orange) leads in every panel; the complete per-cohort results for all six organs are reported in the supplementary (Appendix E, Table 12). On the six-organ HEST panel MIST reaches 0.422 versus 0.398 for BLEEP, the strongest baseline on the six-organ panel. Together, these results show that MIST’s inductive biases carry over to unseen anatomy, which supports RQ2; a complementary mouse experiment in Appendix G shows the same trend.

Key Finding (RQ2): The pooled→LOOO drop, from 0.599 with the organ seen (RQ1) to 0.422 with the organ held out, isolates unseen-organ transfer as the dominant remaining gap; predicting seen organs well does not by itself transfer to a new organ.

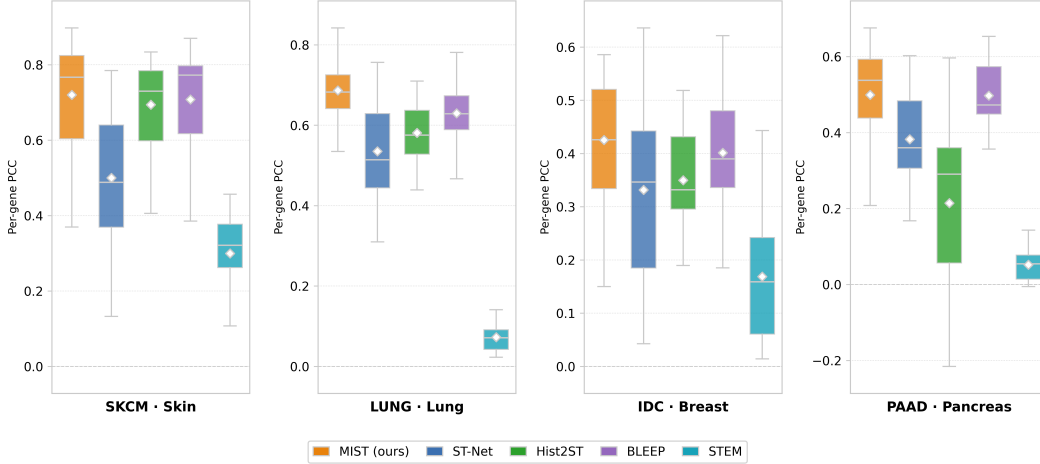


Figure 2: Per-cohort cross-organ LOOO accuracy (four representative held-out cohorts; kidney and prostate omitted for readability, frozen UNI+CONCH). One panel per held-out cohort; box = per-gene PCC over three seeds, $\diamond$ = mean. MIST (orange) leads in all four panels, and wins all six held-out organs in the complete table (Appendix E, Table 12).

5.3 RQ3: What drives accuracy and coordinate invariance?

Given MIST’s accuracy, we next ask which components drive it and whether its coordinate invariance holds in practice. To isolate each context stream, we run a factorial study over all eight combinations of local (L), global (G), and slide-level (S) context on the six-organ pooled set (Table 6). The full model (L + G + S) is best (PCC 0.605), and local attention is the critical stream: adding it (global + local) reaches 0.596, whereas dropping local in favor of global + slide (G + S) falls to 0.570. Crucially, replacing the spatial k NN neighbors with random ones lowers the full model from 0.605 to 0.577, showing that the benefit comes from spatial locality rather than added attention alone. This identifies local spatial-neighborhood aggregation as the primary contributor, which supports the design rationale in RQ3.

Beyond accuracy, we also verify the model’s coordinate invariance empirically. As Table 7 shows, prediction differences under translation and rotation remain below 6×10^{-5} , attributable to floating-point evaluation, and reflection is exact. This numerically confirms the analytic guarantee, completing the answer to RQ3.

Table 7: Numerical coordinate-invariance check. Maximum absolute prediction change under coordinate transforms (random-init model, 257-spot synthetic slide).

Transformation	Max. abs. difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

Intra-cohort spatial gene expression: Skin (SKCM), slide TENX117

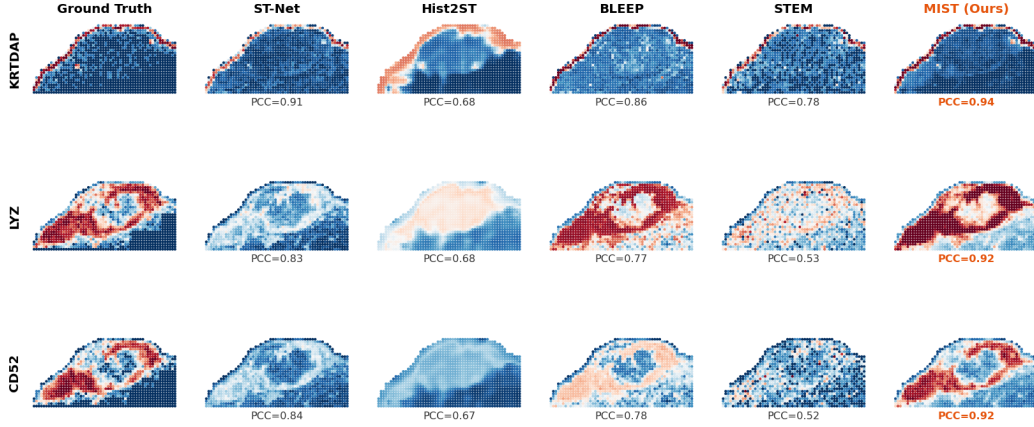


Figure 3: Qualitative spatial expression maps for a held-out skin (SKCM) slide (TENX117), three genes (rows). Each panel colors every spot by expression; MIST (right) most faithfully recovers the ground-truth spatial structure: the perivascular ring for LYZ/CD52 and the epidermal edge for KRTDAP. It attains the highest per-gene PCC (in orange), whereas STEM is noisy and Hist2ST over-smooths. Predictions use identical frozen UNI+CONCH features.

Key Finding (RQ3): Local spatial-neighborhood attention is the primary driver of accuracy (the full L + G + S model scores 0.605, local is the critical stream: global + local reaches 0.596 vs global + slide 0.570, and spatial k NN neighbors beat random ones, 0.605 vs 0.577), and the predictor is provably invariant to spot translation, rotation, and reflection.

Table 6: Context-stream ablation (six-organ POOLED). Local (L), global (G), slide (S) streams; the “Full, random nbr.” row uses random rather than spatial- k NN neighbours. Mean_{std} over three seeds; best per column in bold.

Configuration	Streams	PCC $\uparrow$	MSE $\downarrow$	MAE $\downarrow$
Global + Local	L + G	0.596 $\pm$ 0.007	0.461 $\pm$ 0.024	0.414 $\pm$ 0.010
Global + Slide	G + S	0.570 $\pm$ 0.037	0.517 $\pm$ 0.130	0.439 $\pm$ 0.050
Full, random nbr.	L _{rand} + G + S	0.577 $\pm$ 0.009	0.466 $\pm$ 0.015	0.415 $\pm$ 0.004
MIST (ours)	L + G + S	0.605$\pm$0.006	0.421$\pm$0.011	0.400$\pm$0.013

5.4 RQ4: Spatial structure preservation

Accurate per-gene prediction does not by itself guarantee that predictions preserve coherent tissue regions, so we finally ask whether MIST recovers spatial structure. To test this, we cluster spots (KMeans, $k=6$ domains) from either predicted or measured expression and

compare the induced partitions using the adjusted Rand index (ARI) and normalized mutual information (NMI) on held-out slides of the six-organ subset (Table 8). MIST leads on both ARI (0.173) and NMI (0.242), ahead of the strongest baseline on each metric (BLEEP at 0.147 ARI, ST-Net at 0.201 NMI). Absolute scores vary widely across slides ($\text{std} \approx \text{mean}$), so the cohort means alone are hard to interpret; a paired per-slide comparison, however, is decisive (Figure 4): MIST beats BLEEP on 78% of slides for ARI and 90% for NMI ($n=58$ held-out slides; Wilcoxon $p < 0.001$ for both). MIST therefore recovers spatial domains more faithfully than the strongest baseline, even though the absolute agreement remains modest.

Table 8: Spatial-domain recovery (six-organ subset). ARI/NMI between domains (KMeans, $k=6$) from predicted vs. measured expression; mean_{std} over three seeds; best per column in bold.

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net He et al. (2020)	0.140 _{0.11}	0.201 _{0.12}
Hist2ST Zeng et al. (2022)	0.133 _{0.11}	0.180 _{0.11}
BLEEP Xie et al. (2023)	0.147 _{0.12}	0.197 _{0.13}
STEM Zhu et al. (2025)	0.101 _{0.12}	0.136 _{0.14}
MIST	0.173_{0.14}	0.242_{0.14}

Per-slide spatial-domain recovery ($n=58$ held-out slides)

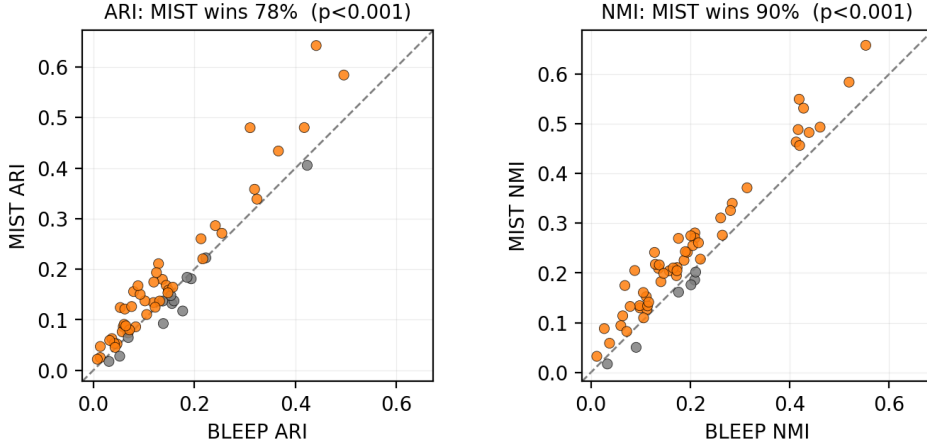


Figure 4: Per-slide spatial-domain recovery, MIST (y) vs. BLEEP (x) on the $n=58$ held-out slides. Each point is one slide; orange marks slides where MIST wins; the dashed line is $y=x$. Points spread widely (large across-slide variance) yet lie mostly above the diagonal, so MIST improves ARI on 78% and NMI on 90% of slides (Wilcoxon $p < 0.001$ for both).

Key Finding (RQ4): MIST reconstructs spatial domains more faithfully than the strongest baseline: paired per slide, it improves both ARI and NMI on 78–90% of held-out slides ($n=58$; Wilcoxon $p < 0.001$; Figure 4), though the absolute domain agreement remains modest.

5.5 Case Study: Recovering Cancer-Marker Genes

Average accuracy is clinically meaningful only if a model recovers the specific genes that define a cancer. We therefore test whether MIST predicts canonical marker genes for two cancers: melanoma (SKCM), through the melanocyte / MITF-pathway markers (MLANA, MITF, TYR, PMEL, DCT, TYRP1, S100B), and breast carcinoma (IDC), through the clinical immunohistochemistry panel (HER2/ERBB2, ER/ESR1, PR/PGR, GATA3, EPCAM,

KRT8). Table 9 reports per-gene Pearson correlation on held-out slides. MIST recovers the melanoma markers almost perfectly (mean PCC 0.896; MLANA 0.948, MITF 0.943, TYR 0.936) and the breast markers strongly (mean 0.716; HER2 0.829), and it beats BLEEP on every marker in both cancers. Consistent with this, Figure 5 shows the predicted spatial maps reproduce the measured tumor architecture ($r \geq 0.91$ on the displayed slides). MIST thus reads out clinically actionable molecular markers, and their spatial localization, from H&E alone.

Table 9: Case study: per-gene recovery of canonical cancer markers. Per-gene Pearson $\uparrow$, mean over folds and three seeds; best per row in bold.

Cancer	Marker	MIST	BLEEP
Melanoma	MLANA	0.948	0.860
	MITF	0.943	0.862
	TYR	0.936	0.841
	S100B	0.888	0.823
	TYRP1	0.868	0.782
	PMEL	0.851	0.768
	DCT	0.842	0.790
	Mean	0.896	0.818
Breast	KRT8	0.881	0.846
	EPCAM	0.871	0.830
	HER2 (ERBB2)	0.829	0.750
	GATA3	0.829	0.786
	KRT5	0.625	0.567
	ER (ESR1)	0.580	0.505
	PR (PGR)	0.395	0.364
	Mean	0.716	0.664

Measured vs. MIST-predicted marker expression (held-out slides; dark = low, bright = high)

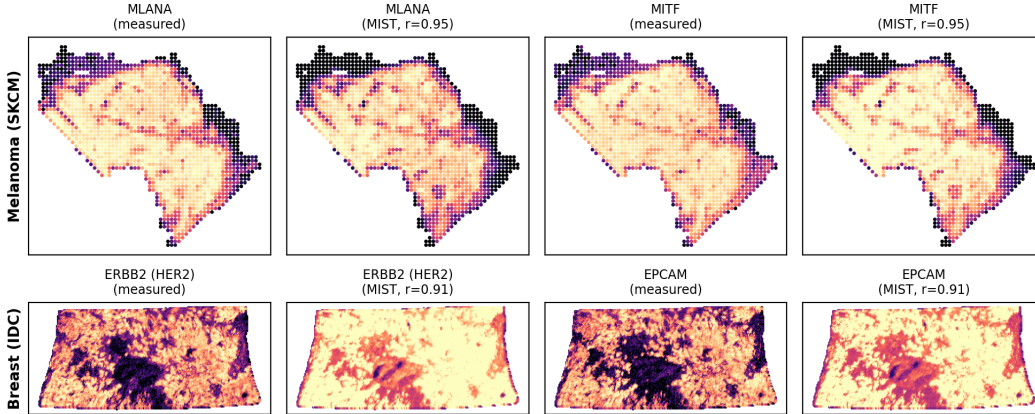


Figure 5: Spatial recovery of cancer markers. Measured vs. MIST-predicted expression on held-out melanoma (top) and breast (bottom) slides for representative markers (dark = low, bright = high). MIST reproduces the spatial architecture of each marker (r shown per panel).

Three findings merit emphasis. First, a compact direct predictor is competitive when features, gene panels, splits, and model selection are controlled; the 11.6% relative gain over the strongest baseline rests on matched experimental conditions, not a larger model. Second, pooled multi-organ performance is not a substitute for unseen-organ evaluation: the drop

from 0.599 (pooled) to 0.422 (LOOO) shows that anatomical shift remains a separate and harder problem. Third, local spatial-neighborhood attention is the dominant contributor in the factorial study; global and slide pathways provide moderate complementary gains, and the complete model (L + G + S) achieves the best mean on the pooled evaluation.

Direct and generative models serve different purposes. STEM [Zhu et al. \(2025\)](#) models a conditional expression distribution and can represent one-to-many uncertainty through iterative sampling. MIST produces a point estimate optimized for correlation and squared error. Its benefits, including deterministic, single-pass inference and explicit coordinate invariance, do not imply that morphology uniquely determines expression.

Limitations. The evaluation predicts 50 genes from frozen UNI+CONCH embeddings; results may not extend to transcriptome-scale output or end-to-end image encoding. LOOO uses a fixed benchmark panel and does not test platform or institution shift simultaneously. Coordinate invariance holds only for the spatial pathway with fixed image features; rotating the histology image can change its frozen embedding. Domain-recovery analysis covers only two baselines and has high slide-level variance. Pooled folds are sample-disjoint but cannot be verified as patient-disjoint where patient identifiers are unavailable.

6 Conclusion and Future Work

We developed MIST, a coordinate-invariant local-global spatial transformer for predicting gene expression from histology. Pairwise-distance encoding makes the spatial pathway invariant to rigid coordinate transformations by construction, while three parallel context signals: local k NN attention, global self-attention, and slide-level pooling, together produce complete-slide predictions in a single pass. These design choices directly address our design goals that multi-scale context is necessary and that architectural positional bias should not break under coordinate shift. Under matched HEST-1k evaluation, MIST improves mean intra-cohort correlation and remains competitive in LOOO transfer; the pooled-to-LOOO gap exposes the remaining difficulty of anatomical transfer as the central open challenge.

For future work, it would be valuable to explore how conditioning on slide-level metadata (organ type, sequencing platform) can narrow this anatomical gap, for instance, through FiLM or adaptive layer normalisation conditioners that are available at inference time without requiring expression labels. Scaling the gene panel beyond 50 genes, incorporating end-to-end image encoding, and combining MIST’s geometric structure with calibrated generative uncertainty modeling (as in STEM) are further natural directions.

AI use statement

Generative AI tools assisted with manuscript organization, language editing, ICLR formatting, and consistency checks between narrative claims and reported tables. They were not used to generate experimental data or execute model experiments. The authors reviewed all AI-assisted revisions and take responsibility for the final content of this work.

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Supplementary Material

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A Dataset

Table 10 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. Two organ groups in HEST-1k contain multiple cohorts: breast (IDC + LYMPH_IDC) and colorectal (COAD + READ). Within each group the constituent cohorts differ substantially in size: IDC has $2.25\times$ more spots than LYMPH_IDC (44,974 vs. 19,964), and COAD has $2.2\times$ more spots than READ (18,523 vs. 8,407). To avoid misrepresentation from unweighted averaging over unequal cohorts, Table 3 uses IDC as the representative breast cohort and COAD as the representative colorectal cohort, discarding the smaller partner cohorts (LYMPH_IDC and READ) from the organ-level rows. This choice maximises statistical reliability of each organ-level estimate. Our cross-organ experiments (leave-one-organ-out and pooled) use the six cohorts marked in Table 10: CCRCC (kidney, 24 samples / 74,220 spots), IDC (breast, 4 / 44,974), LUNG (lung, 2 / 7,505), PAAD (pancreas, 3 / 9,793), PRAD (prostate, 23 / 62,710), and SKCM (skin, 2 / 5,716); COAD, READ, LYMPH_IDC, and HCC are not used in the cross-organ setting.

Table 10: HEST-1k cohort statistics. Spots after barcode filtering; folds = per-cohort train/test splits. The six $\checkmark$ cohorts form the no-COAD cross-organ set; HCC, LYMPH_IDC, and READ are excluded.

Cohort	Organ	Samples	Spots	Folds	6-organ set
IDC	Breast	4	44,974	4	$\checkmark$
LYMPH_IDC	Breast	4	19,964	4	
COAD	Colorectal	4	18,523	2	
READ	Colorectal	4	8,407	2	
CCRCC	Kidney	24	74,220	6	$\checkmark$
HCC	Liver*	2	4,248	2	
LUNG	Lung	2	7,505	2	$\checkmark$
PAAD	Pancreas	3	9,793	3	$\checkmark$
PRAD	Prostate	23	62,710	2	$\checkmark$
SKCM	Skin	2	5,716	2	$\checkmark$

*HCC (liver) is included here for completeness but excluded from Table 3; all baselines score near zero on this cohort due to low spatial gene-expression variance.

Split granularity and patient disjointness. Each “sample” in Table 10 is a distinct HEST-1k slide (a whole-slide section), and every fold is slide-disjoint: no slide is shared between training and test, the pooled regime holds out entire slides, and the leave-one-organ-out regime holds out entire organs. HEST-1k patient/donor identifiers are not distributed with the per-slide data in our pipeline, so for a cohort in which one donor may contribute several slides we cannot verify patient-level disjointness. The intra-cohort and pooled numbers should therefore be read as slide-level generalization and could be optimistic if a cohort mixes multiple slides from the same patient; this risk is largest for the two multi-slide cohorts, CCRCC (24 slides) and PRAD (23 slides), and small for the remaining cohorts (IDC 4, PAAD 3, LUNG 2, SKCM 2 slides). Importantly, the cross-organ LOOO transfer results (our central generalization claim) are unaffected, because an entire organ (and hence a disjoint set of patients) is withheld at test time.

B Extended Related Work

Machine-learning prediction of gene expression from histology. ST-Net [He et al. \(2020\)](#) established patch-level regression. Hist2ST [Zeng et al. \(2022\)](#) adds spatial context through a graph over neighboring spots, while BLEEP [Xie et al. \(2023\)](#) aligns image and expression embeddings for cross-modal retrieval. STEM [Zhu et al. \(2025\)](#) instead treats expression prediction as conditional generation. These methods use different features, gene panels, splits, and selection procedures, so their published scores do not isolate architectural gains; our comparisons therefore use a matched protocol. MIST is a deterministic point predictor and does not model the full conditional expression distribution.

Geometric structure and coordinate invariance. Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this by constructing interactions from relative quantities [Satorras et al. \(2021\)](#). MIST uses pairwise Euclidean distances, which are preserved by translations, rotations, and reflections; the guarantee applies to the spatial pathway for fixed histology features, not to rotations of the image-encoder input.

Generalization across samples and organs. HEST-1k [Jaume et al. \(2024\)](#) spans many cohorts and organs. Within-cohort evaluation can exploit organ-specific morphology or acquisition effects. We therefore distinguish pooled cross-validation, which holds out samples while retaining their organ types in training, from LOOO, which withholds an entire organ and provides a stricter test of anatomical transfer.

Positioning of MIST. MIST connects these three directions: an inductive-bias-explicit spatial transformer for direct regression, evaluated under a matched cross-cohort protocol. Unlike absolute position encodings or frame-averaging strategies, its coordinate pathway is invariant by construction. Unlike generative models, it does not represent expression uncertainty; its advantage is deterministic, single-pass inference with explicit geometric structure. Unlike prior evaluations that report within-cohort accuracy only, our controlled comparison separates interpolation from anatomical transfer, making the difficulty of organ shift directly measurable.

C Extended Experimental Setup

Dataset and preprocessing. We evaluate on HEST-1k [Jaume et al. \(2024\)](#) using the cohorts of Table 10; HCC, LYMPH_IDC, and READ are excluded under the documented quality and split criteria, and the six cohorts marked in that table (CCRCC, IDC, LUNG, PAAD, PRAD, SKCM) form the cross-organ set. We use the provided patient-stratified folds, $\log(1 + x)$ normalization, and frozen UNI+CONCH features (1536-d). Unless stated otherwise, targets are the 50-gene HEST HVG panel. The panel study additionally uses matched 50-gene DEG (Wilcoxon selection across Leiden clusters) and HMMVG (mean-and-variability selection after variance filtering) panels (Table 2).

Evaluation protocol. Intra-cohort training fits a model within each cohort. Pooled five-fold validation trains across organs while holding out slide-level samples; organ identity defines the protocol but is not a model input. LOOO trains on all-but-one organ and tests the held-out organ. We report PCC, SCC, and MSE averaged over outer folds and three seeds. Checkpoints are selected on inner validation and evaluated once on the outer test fold; single-slide folds use disjoint spots for inner validation. Patient-level disjointness is not guaranteed where identifiers are unavailable.

Baselines and implementation. We compare with ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) under matched features, panels, splits, budgets, and checkpoint selection. MIST uses four layers, width 256, four heads, eight neighbors, 16 radial bases, and 4.88M parameters. AdamW training uses learning rate 10^{-3} , weight decay 10^{-2} , cosine decay, gradient clipping, 100 epochs maximum, patience 20, and at most 3,000 spots per step.

D Encoder Ablation

We repeat the intra-cohort PCC@50 comparison (the 50-gene block of Table 3, which uses the concatenated UNI + CONCH features) with each foundation encoder in isolation: UNI only (ViT-L/16, 1024-d) and CONCH only (512-d), holding models, cohorts, gene panels, splits, and training budget fixed (Table 11). MIST remains the most accurate encoder-averaged method under both UNI-only and CONCH-only features.

Table 11: Encoder ablation: intra-cohort PCC@50 per encoder. UNI-only (1024-d) vs. CONCH-only (512-d); mean_{std} over three seeds; best per row in bold.

Encoder	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
UNI (1024-d)	IDC Breast	0.538±0.008	0.250±0.067	0.508±0.005	0.436±0.005	0.569±0.009
	CCRCC Kidney	0.172±0.010	0.214±0.015	0.184±0.007	0.116±0.002	0.264±0.010
	LUNG Lung	0.482±0.012	0.378±0.097	0.559±0.007	0.393±0.009	0.590±0.005
	PAAD Pancreas	0.353±0.022	0.243±0.061	0.393±0.035	0.227±0.017	0.392±0.040
	PRAD Prostate	0.251±0.008	0.322±0.020	0.343±0.009	0.190±0.010	0.394±0.003
	SKCM Skin	0.632±0.010	0.510±0.015	0.639±0.015	0.471±0.031	0.694±0.001
	Average	0.405	0.319	0.438	0.305	0.484
CONCH (512-d)	IDC Breast	0.515±0.007	0.264±0.018	0.496±0.011	0.396±0.011	0.551±0.004
	CCRCC Kidney	0.160±0.010	0.126±0.026	0.141±0.012	0.078±0.001	0.222±0.010
	LUNG Lung	0.517±0.001	0.435±0.006	0.576±0.003	0.386±0.030	0.554±0.005
	PAAD Pancreas	0.372±0.035	0.186±0.035	0.384±0.033	0.251±0.028	0.387±0.045
	PRAD Prostate	0.327±0.003	0.322±0.009	0.364±0.010	0.210±0.013	0.406±0.006
	SKCM Skin	0.633±0.002	0.388±0.022	0.633±0.005	0.453±0.014	0.658±0.007
	Average	0.421	0.287	0.432	0.296	0.463

E Complete LOOO (Leave-One-Organ-Out) Results

Table 12 reports the complete per-cohort leave-one-organ-out transfer results on the six-organ subset summarised in the main text (Section 5.2, Figure 2). Each fold holds out one entire organ and trains on the other five, using leakage-safe per-fold gene panels and identical frozen UNI + CONCH features. MIST wins every held-out cohort and the average (0.422).

Table 12: Cross-organ LOOO transfer (six-organ subset). Per-cohort Pearson ↑; each fold holds out one organ (leakage-safe panels, frozen UNI + CONCH); mean_{std} over three seeds; best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25 [†]	MIST (Ours)
CCRCC Kidney	0.074±0.00	0.064±0.01	0.058±0.00	0.024±0.01	0.098±0.01
IDC Breast	0.332±0.02	0.350±0.00	0.401±0.04	0.131±0.08	0.425±0.03
LUNG Lung	0.535±0.07	0.581±0.00	0.630±0.00	0.067±0.00	0.687±0.02
PAAD Pancreas	0.382±0.00	0.214±0.17	0.497±0.03	0.052±0.00	0.499±0.03
PRAD Prostate	0.078±0.01	0.062±0.01	0.091±0.00	0.016±0.00	0.105±0.01
SKCM Skin	0.500±0.09	0.694±0.02	0.708±0.01	0.299±0.00	0.720±0.03
Average	0.317	0.328	0.398	0.098	0.422

[†]STEM incomplete (1–2 of 3 seeds); values provisional.

F Computation Resources, Model Complexity, and Inference Scaling

All experiments ran on a shared GPU cluster. Feature extraction and training used NVIDIA GPUs with compute capability $\geq$ sm_80 (A100 80GB, A100-PCIE, L40s, and H200); Volta (V100) GPUs are incompatible with our CUDA-13 / PyTorch build and were excluded.

- Encoders: frozen UNI (ViT-L/16, 1024-d) and CONCH (512-d) patch features, concatenated to 1536-d; no encoder fine-tuning.
- Model: MIST has 4.88M trainable parameters (Table 13) and predicts a whole slide in a single forward pass, with no iterative sampling.
- Training: AdamW with a cosine schedule, 100 epochs with early stopping (patience 20), nested validation, and a correlation-aware objective; three seeds per configuration.

- Inference: single-pass; median model-only latency and peak memory versus slide size are reported in Table 14, with feature extraction and disk I/O excluded.

Table 13: Model complexity. Parameter counts and inference mechanisms.

Model	#Params (M)	Inference
ST-Net He et al. (2020) (NBE’20)	0.47	single-pass
Hist2ST Zeng et al. (2022) (BIB’22)	2.18	single-pass
BLEEP Xie et al. (2023) (NeurIPS’23)	0.54	single-pass
STEM Zhu et al. (2025) (ICLR’25)	34.16	iterative
MIST (V3)	4.88	single-pass

Table 14: MIST inference scaling (A100 80GB). Median model-only latency and peak GPU memory vs. slide size; feature extraction and I/O excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

G Mouse cross-organ diagnostic

We evaluate 59 mouse Visium slides spanning seven organs with mouse-specific 50-gene panels. Pooled performance reaches 0.887, whereas every method remains near the floor under LOOO; we therefore do not interpret the latter as evidence of successful cross-species organ transfer.